Supporting Information

S1. Study evidence hierarchy and artifact lineage

The study contains four evidence streams that should not be pooled: historical Stage 3 context; the WUR-v2 grouped model-development program; the single frozen external MSnLib evaluation; and a separate 33-compound collision-energy-interface Design A study. The primary MSnLib result comes from the frozen release bfa153e828e08b3c7e674ef7fe9d6818d14a2b61. The Phase 4 manuscript and derived figures do not alter the original experimental outcomes.

Source artifacts include artifacts/wur_v2_confirmation_v2/result/measured_mu.csv, external_analysis.json, the frozen candidate and reference model records in artifacts/wur_v2, the source analysis script scripts/wur_v2_confirmation_v2/11_external_analysis.py, the comparator result at artifacts/comparator_benchmark/result/analysis.json, and Design A result at artifacts/ce_interface_adjudication/design_a/result/analysis.json. The complete repository release manifest supplies hashes. Tables in COMPLETE_SUPPLEMENTARY_TABLES/ identify the source JSON or CSV for each export.

S2. Population flow, inclusion, and exposure history

Stage	Compounds	Notes
MSnLib measured population	1,794	Each has measurements at two numeric settings (20 and 60); selected converted mzML scans annotate them in electronvolts, but native methods and RAW linkage remain unverified
Outside candidate/reference profile support	5	Excluded if either model was unsupported at either rung
Primary scored population	1,789	Two compound-rung cells per compound; 3,578 cells
Scaffold groups	1,687	Bemis–Murcko grouping; largest group size 7
Comparator common population	1,327	1,254 scaffold groups; a distinct benchmark population
Comparator source candidates removed	0	No excluded predicted/measurement cases in the frozen primary comparator output

A prior MSnLib sample was exposed during parser preflight before the planned outcome analysis. It was not scored and was permanently excluded. The replacement sample was selected once under a frozen outcome-blind protocol after excluding recorded exposed compound keys and scaffolds. That control does not establish that every close analogue was excluded or that no unrecorded exposure occurred. The replacement population was evaluated once and is now exposed. It cannot support a second confirmatory use.

WUR-v2 development used 1,325 positive-mode compounds and a five-fold scaffold-disjoint outer evaluation; 13 model-experiment families were documented. The chosen candidate was selected by a simplicity rule among near-best models. The MSnLib primary analysis is separate from development selection. Exact sample lineage and exclusions are defined in the frozen protocol and exposure registry.

S3. Measured endpoint and data processing

For a processed spectrum with masses m_j, raw intensities $I_j \geq 0$, and declared protonated precursor mass m_prec, the endpoint is:

$\mu = (\Sigma_j\ I_j\ m_j) / (m_{prec}\ \Sigma_j\ I_j)$.

The implemented base preprocessing has zero relative-intensity cutoff, includes the precursor peak, and uses raw peak intensities. The per-spectrum function checks finite positive m/z, nonnegative intensities, positive total signal, and positive precursor mass. Duplicate compound-energy scans are reduced to one

measured value by the frozen median aggregator. The WUR development path uses the MassBank declared PRECURSOR_M/Z; the MSnLib validation path passes structure-derived [M+H]+ to the endpoint. The same formula is applied, but the denominator source differs and equality is not assumed. The WUR numerator also includes observed peaks, so the endpoint can slightly exceed one. The endpoint is dimensionless and nonnegative, but can be slightly greater than one if observed peaks are marginally heavier than the declared precursor. It is not a count of cleaved bonds, precursor survival yield, fragment identity, mechanism, or complete-spectrum similarity.

The formula is implemented by src/muru/features.py::mu after Spectrum.preprocess in src/muru/io/wur_spectra.py::spectrum_mu; base settings are defined in src/muru/wur_bridge_constants.py. No inference is made from the scalar endpoint about fragment identity or pathway.

S4. Molecular representation and model fitting

Component	Frozen setting
Compound parent	Largest organic fragment; neutralized where a neutral form exists; permanent cations retain charge
Stereochemistry	Removed from model feature structure
Identity key	First InChIKey block
Tier A	12 fixed physicochemical/molecular descriptors, divided by descriptor-specific constants
Candidate Tier A scaling	Further standardized using training-fold mean and SD
Reference Tier A scaling	No additional fold-specific standardization
Morgan fingerprint	RDKit 2026.03.5; radius 2; 2,048 hashed count bins; no chirality; log1p
Candidate feature block weight	Morgan block multiplied by 0.1
Candidate ridge	α=0.3; search $\alpha \in \{0.1,0.3,1,3,10,30,100,300\}$, Morgan weight$\in\{0.1,0.3,1.0\}$
Tier A reference ridge	α=0.1; search $\alpha \in \{0.01,0.1,1,10,100\}$
Ridge intercept	Fitted and unpenalized (fit_intercept=True)
Profile	Shared decreasing isotonic response; 60 knots on log-u
Profile/scale fitting	Three alternating updates; log-scale grid of 241 points on [−1.6,1.6], followed by parabolic refinement
Model population	1,325 compounds in development records

Let x_i be the model-specific feature vector, $\ell_i=\log(g_i)$ the target log-scale, w_i the normalized curvature weight, and α the ridge penalty. The implemented weighted ridge objective is:

$$(\hat{b}_0,\hat{\beta})=\operatorname{argmin}_{(b_0,\beta)} \Sigma_i\, w_i(\ell_i-b_0-x_i^T\beta)^2 + \alpha\|\beta\|^2.$$

The intercept b_0 is not penalized. Candidate and reference use the same shared response profile and scale parameterization, but they are separately selected pipelines with different feature spaces, candidate descriptor scaling, dimensionality, and α grids. Their contrast cannot isolate a Morgan-fingerprint effect.

S5. Response profile and external energy input

The response profile is written $\Phi(u)$, with normalized coordinate $u=(E/30)/g$; the predicted endpoint is $\hat{\mu}_i(E)=\Phi((E/30)/\hat{g}_i)$. The profile is a decreasing isotonic fit represented by 60 fixed knots. Development fitting uses the selected training data only.

The WUR primary-library publication/acquisition context describes HCD normalized settings 15, 30, 45, 60, 75, 90 and stepped SCE settings 25, 38, 59; the exported collision-energy field itself is numeric and unitless. The 10-ppm peak-matching tolerance applies to auxiliary precursor matching and does not change μ. For the selected MSnLib scans, Phase 6 verified converted mzML collision-energy terms with electronvolt units against allowlisted source-member hashes [12–14]. The v5 repository describes the converted ID-X collection [13], and the ID-X manual distinguishes normalized and absolute settings [15]. A separate search found 161 .meth files in four archives in the public RAW-positive record [33]; they are

Vanquish_H LC SeparationMethod files with no explicit ID-X/NCE/CE setting or run link and therefore do not establish the MS method. However, native ID-X method files, conversion logs, and exact RAW-to-mzML checksum links remain unavailable; the original protocol called the values NCE. We therefore state that converted mzML encodes eV while native setting and physical energy comparability remain unverified. The development bridge maps LCSB nominal coordinate E_LCSB to the WUR coordinate using T(E_LCSB)=a+bE_LCSB, where a=−5.95552603907965, b=0.8618030610784555; external prediction applies its inverse to the recorded values. The adapter is reproducible but is not a physical conversion or cross-instrument calibration. Under A0 the protocol assumed ID-X NCE equals IQ-X NCE; anchor absolute-agreement thresholds failed at A0 (median |Δμ| 0.0501 vs 0.05), A1 (pooled RMSD 0.0811 vs 0.08), and A2 (median |Δμ| 0.0506).

Design A evaluated three comparator energy conventions on 33 compounds. It used cosine similarity as primary metric and paired molecule-level bootstrap because all 33 scaffold groups were unique. K1 had the highest score, 0.5570699532, but adjusted K1−K2 and K1−K3 intervals were [−0.0126596, 0.0990831] and [−0.0074389, 0.1099934]. The frozen conclusion is INTERFACE_UNRESOLVED. The full estimates and source record are exported in COMPLETE_SUPPLEMENTARY_TABLES/ce_design_a.csv.

S6. Metrics and uncertainty

For compound i, rung E, and prediction error $e_iE=\hat{\mu_i}(E)-\mu_i(E)$, pooled P1 is:

$P1 = \sqrt{(\Sigma_i \Sigma_E\ e_iE^2 / N_cells)}$.

Each of the 1,789 scored compounds contributes two cells. The candidate/reference ratio is P1_candidate/P1_reference; values below one favor the candidate. Compound-level summaries include the mean and median compound RMSE, Q90, Q95, CVaR95 (mean of the highest 5% of compound RMSE values), AF (fraction with compound RMSE >0.20), and AF_max (fraction with any rung absolute error >0.30). The AF and AF_max thresholds are frozen engineering thresholds, not physical constants.

The primary 95% interval is a paired scaffold bootstrap with 10,000 resamples and seed 20261011. Each replicate draws 1,687 scaffold groups with replacement and carries all compounds and both rung cells in each sampled group. P1 for both models and their ratio are recomputed per replicate. The interval uses empirical 2.5th and 97.5th percentiles. It quantifies sampling uncertainty across the observed scaffold groups conditional on the frozen models, eligible sample, endpoint, and energy map. It does not include training or model-selection uncertainty.

S7. Full primary external results

Exact source values are available in COMPLETE_SUPPLEMENTARY_TABLES/primary_results.csv, per_energy_results.csv, error_summary.csv, external_compound_errors.csv, and novelty_results.csv.

Result	Value
Candidate P1	0.1296115107
Tier A reference P1	0.1335276112
Candidate/reference ratio	0.9706719801
95% scaffold bootstrap interval	[0.9523611880, 0.9892672062]
Relative P1 reduction	2.9328%
Prespecified practical target	ratio ≤0.95
Target passed?	No

Rung	n	Candidate RMSE	Reference RMSE	Ratio	Candidate mean residual	Reference mean residual
Recorded setting 20 (converted mzML eV	1,789	0.1643762	0.1673566	0.9821913	−0.0077006	+0.0353532

annotation)						
Recorded setting 60 (converted mzML eV annotation)	1,789	0.0811095	0.0874700	0.9272829	+0.0033380	+0.0206737

S8. Novelty, error distribution, and tail outcomes

Frozen novelty quartile	n	Candidate P1	Reference P1	Ratio
Q1, lowest maximum similarity	449	0.12768	0.12663	1.0082968
Q2	462	0.11787	0.12208	0.9655334
Q3	456	0.12356	0.12880	0.9592956
Q4, highest similarity	422	0.12686	0.13316	0.9527711

Novelty bins use maximum Morgan-count Tanimoto similarity to any of 1,325 development compounds. Quartile estimates are descriptive and no quartile-specific uncertainty intervals were estimated. The most-novel quartile has slightly worse candidate point error.

Compound-level measure	Candidate	Reference
MRMSE	0.1121374	0.1175158
Median RMSE	0.0995566	0.1075164
Q90	0.2070537	0.2065879
Q95	0.2315082	0.2348610
CVaR95	0.2656786	0.2648635
AF	0.1185020	0.1179430
AF_max	0.0620458	0.0609279

The scaffold-bootstrap AF difference was +0.0005590, 95% interval [−0.0145253,+0.0155383]. Tail-risk benefit was not demonstrated; these data also do not prove equality or harm.

S9. Comparator benchmark under both tested energy conventions

The comparator study used 1,327 common compounds and 1,254 scaffold groups. Its primary input condition was native NCE for FIORA-OS and ev_primary for ICEBERG and GLACIER; only the latter two have a raw-NCE diagnostic. It converted predicted fragment peak lists from each full-spectrum system to the precursor-including μ endpoint with the frozen spectrum_mu function and calculated the same two-rung pooled P1. Thus the comparison uses a common derived scalar endpoint. It does not score spectrum similarity, peak identity, candidate identification, or complete-spectrum fidelity. The study is not feature-matched; systems differ in training data, architecture, preprocessing, tuning, and energy-input semantics.

Model	Comparator P1	MURU/comparator ratio (mapped primary)	Ratio CI (95%)	Raw-NCE ratio	Interpretation
FIORA-OS v0.1.0	0.2045525	0.6185250	[0.5933494,0.6456244]	not calculated	Conditional μ comparison
ICEBERG 2.1	0.1620766	0.7806238	[0.7479332,0.8145965]	1.076	Conditional; raw-NCE point ordering reverses
GLACIER	0.1383865	0.9142574	[0.8729690,0.9584434]	1.238	Conditional; raw-NCE point ordering reverses

The mapped primary outputs showed lower MURU μ error for this benchmark. Under the raw-NCE diagnostic, ICEBERG and GLACIER point ratios exceed one. These are convention-dependent benchmark results and do not support a universal full-spectrum ranking. The output and exact CIs are source-derived in comparator_metrics.csv.

S10. Development evidence and unresolved max-T provenance

The development point estimates were P1=0.1159994 for the selected candidate and 0.1305460 for the Tier A reference, ratio 0.8885713. The development protocol amendment A-3 restated the c4 permutation rule after development outcomes and five adversarial reviews were visible but before external outcome access. The literal rule failed for EXP08B (three of five corrected permutation ratios were below 1); the restated rule treated the controls as tests of whether a permuted fingerprint block outperformed the unpermuted candidate and the empty-block null. The five permuted candidates lost to EXP08B by about 11%, with average ratio 1.0003 against null ratio 0.9989. This amendment did not change selected candidate or frozen outputs, but it is a post-results change in the admission interpretation and weakens the prospective status of the development selection. The release reports a ten-arm simultaneous max-T interval [0.860,0.918]. The recovered Phase 2 report records a max-T analysis but not the original script, exact seed, serialized resamples, or selection ledger. Phase 2 also contains a distinct conditional interval [0.8650671579,0.9105586626], which is not interchangeable with the simultaneous interval. Neither value is presented as independently reproduced in Phase 4. The external bootstrap is a different inferential calculation and does not repair the development provenance gap.

S11. Census neutral-scaffold defect erratum

The recovered Phase 2 code called neutral_scaffold(group_key, smiles) although its function signature is neutral_scaffold(smiles, key). Reconstructing the historical and corrected sets yielded 2,090 and 766 neutral-scaffold labels, respectively. The Phase 2 verifier reran the population membership chain for four recorded frames; Phase 4 reran the same verifier read-only from the recovered source and input worktrees.

Population frame	n	Frozen population SHA-256	Historical/corrected membership symmetric difference
DESIGN, step 10b	39,238	8ae32fb53e27a0fb99f875f48d7ffbe127fc1668e512b65797c3ad70712cd09e	0
DESIGN, step 12b	47,742	84828abfe29e595379baceb993e97edc846268f255935a66b675c59407c1e283	0
DETECTED_POSITIVE, step 10b	32,154	feaade612cb35cde6c627b813b9dc69a9dae966ed1bc05b7fd6ba0af9741e451	0
DETECTED_POSITIVE, step 12b	39,236	b526ca0b9b93c7dc39f8b463d2b77f40541b44b0b301e7e3bdacebfc52737aca	0

The rerun verified all four historical reconstructions against frozen counts and hashes and found the watched inputs unchanged. This establishes no membership impact for these four frames only. It cannot establish that all unrecorded intermediate tables, manual decisions, or downstream calculations were unaffected. No frozen input was changed. The read-only verifier short-circuited a strict-similarity diagnostic that cannot affect membership at steps 10b and 12b. The machine-readable sanitized run record is ../verification/census_neutral_filter_verification_rerun.json.

S12. Software, rights, reproducibility, and limitations

The recovered Phase 2 reconstruction record lists Python 3.13.12, NumPy 2.5.2, SciPy 1.18.0, pandas 3.0.5, scikit-learn 1.9.0, RDKit 2026.03.5, and pymzML 2.6.1. Phase 4 figure regeneration uses Python 3.13 and Matplotlib 3.11.1 with NumPy/pandas in the local environment; exact runtime snapshots and hashes are recorded in the release manifest.

The reviewer archive supports statistical reconstruction from committed measured-μ values and frozen model output. It does not include third-party MSnLib raw mzML archives, and does not claim raw-data-to-model reproduction or training from scratch. A system Python environment without RDKit failed to run the census verifier; the preserved Phase 2 worktree environment ran it successfully. Third-party raw files must be acquired under their source terms. Rights-cleared project code and derived tables remain in the project repository.

No human or animal subjects were involved. The principal scientific limits are the scalar target, positive-mode HCD data, two external settings whose converted mzML records encode eV but whose native methods and cross-instrument comparability remain unresolved, profile-support exclusions, the single exposed replacement sample, the physical energy-map ambiguity, the small external effect, the non-improvement in the least-similar quartile, no demonstrated tail advantage, and unreproduced development max-T uncertainty provenance.

S13. Public acquisition-method file search

Phase 6 searched all nine evaluated mzML archives, the seven-family public positive RAW record, and the MassIVE MSV000094528 file listing [34]. The evaluated mzML ZIPs contained no native methods. The Zenodo archive had 161 CRC-verified Thermo .meth files; all were Vanquish_H / SeparationMethod records without an explicit ID-X, NCE, or collision-energy value and without a selected-run link. These are LC methods, not Orbitrap HCD methods.

The MassIVE listing contained 18,281 entries: 9,183 RAW and 9,085 mzML. It contained no .meth, .method, .mth, .msmethod, or .rawmethod extensions. Normalized filename matching produced 835 candidate associations among 1,428 mzML sourceFile names. These candidates do not establish identity. One MCEBIO RAW file (31,994,595 bytes) was streamed and its SHA-1 matched the mzML sourceFile checksum (ea7e6f809882e4a9ff4c04f7cef2011ca2c6efb3), establishing exact file identity for one pair only. A distinct filename-associated Zenodo RAW object mismatched; those repository files are not treated as the same object.

Public archive family	.meth members	Parsed method type	Relevance to ID-X MS collision energy
MCEBIO	67	Vanquish_H / SeparationMethod	No explicit ID-X energy setting or run link
NIHNP	54	Vanquish_H / SeparationMethod	No explicit ID-X energy setting or run link
MCESCAF	1	Vanquish_H / SeparationMethod	No explicit ID-X energy setting or run link
MCEDRUG	39	Vanquish_H / SeparationMethod	No explicit ID-X energy setting or run link
OTAVAPEP, ENAMMOL, ENAMDISC	0	None listed	No method file in listed archive
MCEDIV-20K, TARGETMOL-NPHTS	0	Raw family absent from public raw-positive record	No raw method evidence in that record

The nested NIH nih_positive_mzml.7z member had a verified outer ZIP size of 1,040,150,902 bytes and CRC-32 8d41051a; its listing contained 535 .mzML names and one directory entry, with no method-like extension. Only the listing was inspected. These archive checks narrow the search but do not authenticate a native method or source-to-run identity beyond the single exact RAW/mzML file match. Detailed per-member summaries and hashes are retained in the Phase 6 metadata directory; binary method payloads are not distributed.